SOME NEW EQUATIONS TO DESCRIBE PHOSPHATE SORPTION BY SOILS

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Summary

Phosphate sorption was studied on surface and subsurface soils sampled from the old field experiments at Askov, Denmark. Two empirical three-parameter equations, an extended Freundlich and an extended Langmuir equation are proposed and compared with the Langmuir, the Freundlich, the 'double' Langmuir, the Gunary, and the Fitter-Sutton equations.

In the proposed equations the affinity parameter of the Langmuir equation and the exponent of the Freundlich equation are replaced by the term Bc^{-D} the value of which decreases with increasing phosphate concentration, c. On average the Freundlich equation thus modified yielded the closest fit to the sorption data. This was followed by the Fitter-Sutton, the modified Langmuir, the 'double' Langmuir, the Gunary, the Freundlich, and finally the Langmuir equation. Out of the three equations that yielded the closest fit to the sorption data the correlation between the parameters within the equations, furthermore, was least for the proposed extended Freundlich equation. Therefore, this equation may be generally suitable for describing phosphate sorption by soils.

Introduction

BARROW (1978) gave two motives for studying sorption equations, first to understand the processes involved, and second to summarize many results in a few numbers.

The objective of this paper is primarily concerned with the latter. Two three-parameter sorption equations, an extended Freundlich and an extended Langmuir equation are proposed and compared with five equations from the literature.

Theory

Reviews of phosphate (P) sorption by soils have been given by Larsen (1967), Barrow (1978), and Parfitt (1978). As it is difficult to distinguish between true adsorption and precipitation by means of sorption curves (Veith and Sposito, 1977), the term sorption is used here.

The most popular equations for describing P sorption by soils have been the Langmuir equation:

$$m = \frac{ABc}{1 + Bc} \tag{1}$$

and the Freundlich equation:

$$m = Ac^{B}$$

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where m is the amount of P sorbed per unit weight of soil, c is the concentration of P in the solution, and A and B are constants. In both equations, A may be thought of as governing the position of the curves, while B may be thought of as governing the shape of the curves.

The parameters of the Langmuir equation have often been considered to have a specific significance with A as the sorption maximum and B as an equilibrium constant related to the bonding energy of the P sorbed. The model, however, assumes a constant bonding energy irrespective of the amount of P sorbed, an assumption which according to Bowden *et al.* (1977) does not hold.

The Freundlich equation originally was empirical, but it can be derived theoretically by assuming that the bonding energy decreases logarithmically, as the fraction of surface covered increases.

The two equations have been widely used to describe P sorption but with limited success in experiments covering a wide P concentration range (Olsen and Watanabe, 1957; Gunary, 1970; Bache and Williams, 1971; Syers *et al.*, 1973; Holford *et al.*, 1974). This has led to use of extended equations.

Shapiro and Fried (1959), Syers *et al.*, (1973) and Holford *et al.*, (1974) used the 'double' Langmuir equation:

$$m = \frac{A_1 B_1 c}{1 + B_1 c} + \frac{A_2 B_2 c}{1 + B_2 c} \tag{3}$$

assuming that their soils contained two kinds of P sorbing sites. Gunary (1970) added a square root term to one of the linear forms of the Langmuir equation. Isolating m gives the following equation:

$$m = \frac{1}{A + B/c + \mathrm{D}/\sqrt{c}} \tag{4}$$

The reciprocal of the constant A was designated as the phosphate sorption maximum, but no interpretation of the other terms was offered.

Fitter and Sutton (1975) extended the Freundlich equation by including a constant, D which should account for the phosphate already present in the soil:

$$m = Ac^B - D \tag{5}$$

However, soils are complex with surfaces of mostly unknown heterogeneity. It is unlikely that the P is sorbed under such well-defined conditions as the above mentioned equations require. Therefore, the equations are best treated as empirical equations describing actual sorption processes.

The proposed two equations:

$$m = \frac{ABc^{-D}c}{1 + Bc^{-D}c} \tag{6}$$

$$m = Ac^{Bc^{-D}} \tag{7}$$

are also empirical. These are respectively extended Langmuir and Freundlich equations where the 'shape governing' parameter, B of the

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original equations is replaced by a 'shape governing' term:

$$Bc^{-D}$$

the value of which varies with the P concentration, c. D is a constant.

Soils and Methods

The soils were sampled from two sites, Lermark and Sandmark, of the old field experiments at Askov Experiment Station, Denmark (Iversen and Dorph-Petersen, 1951). From each site surface and subsurface samples were taken from a plot treated with FYM and from a plot treated with commercial fertilizers. Some properties of the soils are given in Table 1. The soils were air-dried and passed through a 1 mm sieve. P sorption data were obtained by shaking 5 g of soil with 100 ml 0.01 M $CaCl_2$ containing KH_2PO_4 at 15 concentrations ranging from 0 to 1200 μM P. Three drops of chloroform were added, after which the suspensions were shaken end over end (30 rpm) for 24 h, at 25 °C. The suspensions were centrifuged, and the centrifugates were analyzed for P.

The quantity of P sorbed during the shaking was calculated as the difference between the initial and final quantities of P in the solution. The original P already present in the soil was estimated by anion-exchange resin extraction for 24 h using bicarbonate saturated resin enclosed in mesh bags (Sibbesen, 1977, 1978).

Two data sets of total P sorbed were created:

1) P sorbed during the experiment + resin extracted P.
2) P sorbed during the experiment + 1.5 times resin extracted P.

TABLE 1
Some properties of the soils used

No.	*Origin and soil type*	*Nutrients added for 80 years, kg/ha per year*	*Depth, cm below surface*	*pH $CaCl_2$*	*P resin µmol P/g soil*
1a	Askov Lermark, sandy loam	105K 26P 100K in fertilizer	0–20	6.2	0.98
1b			20–40	6.0	0.23
2a		139N 29P 87K in FYM	0–20	6.0	1.48
2b			20–40	5.9	0.40
3a	Askov Sandmark, sand	70N 18P 66K in fertilizer	0–20	5.7	1.83
3b			20–40	5.7	1.68
4a		93N 20P 58K in FYM	0–20	5.8	1.61
4b			20–40	5.5	1.74

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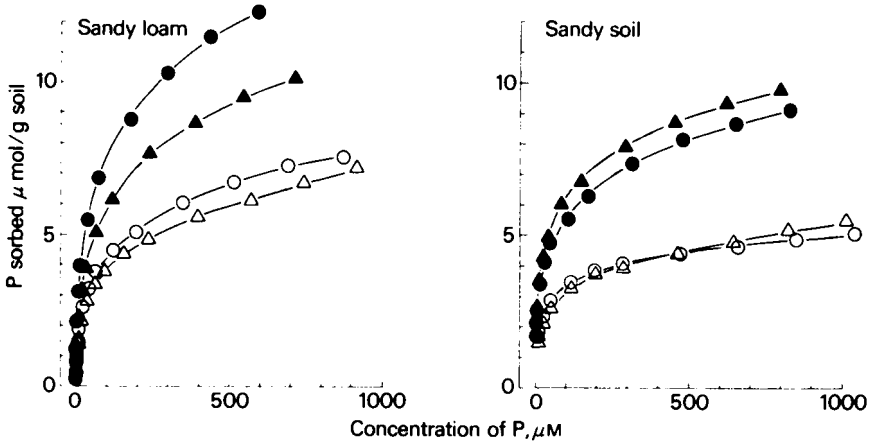

FIG. 1. Phosphate sorbed by soil samples taken from a sandy loam field and a sandy soil field of an old field experiment. ○ from plough layer of plots treated with commercial fertilizers (1a, 3a). ● from subsurface of plots treated with commercial fertilizers (1b, 3b). △ from plough layer of plots treated with farm yard manure (2a, 4a). ▲ from subsurface of plots treated with farm yard manure (2b, 4b).

The seven mathematical equations mentioned above were fitted to the non transformed data of total P sorbed by the use of a SAS computer programme for non-linear regression analysis (Barr *et al.*, 1976). The programme produces least squares estimates of the parameters of non-linear models by an iterative process.

Results and Discussion

The P sorption curves in Fig. 1 described by the proposed extended Freundlich equation (7) show that (a) the subsurface samples sorbed more P than the corresponding surface samples, (b) the curves for the farm yard

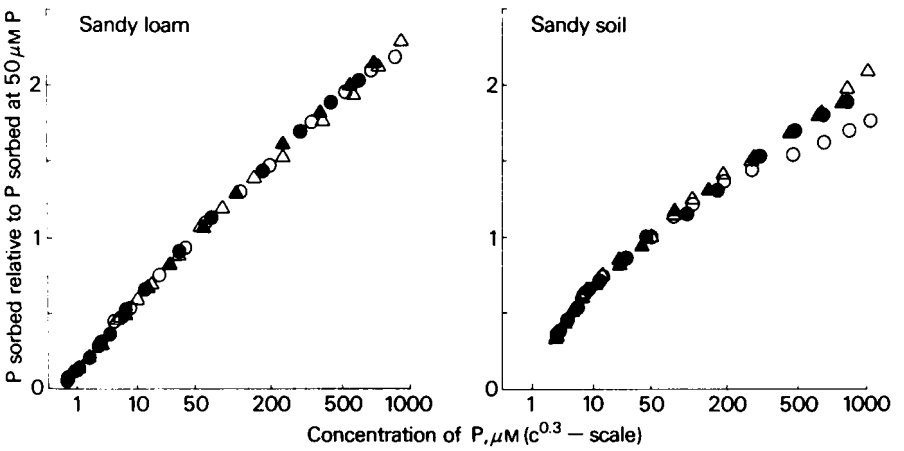

FIG. 2. Data of Fig. 1 replotted on transformed scales.

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manured surface samples (2a, 4a) tended to be more linear in the upper part than the curves for the corresponding surface samples that received commercial fertilizers (1a, 3a).

A better visual comparison of the type of the curves is obtained from Fig. 2, where P sorbed is plotted on a unity scale versus P concentration raised to a power of 0.3. It may be seen that the curves for the sandy soils (3, 4) curved more to the right than the sandy loam curves (1, 2) indicating the lower affinity for P by the sandy soils. The curves for three of the sandy loam samples, 1a, 1b, and 2b coincided indicating the same affinity for P of the three samples. They differed only in the quantity of P sorbed.

The ability of the seven equations to describe data set 1 is shown in Table 2, which gives the unexplained variation relative to the total variation.

Of the two-parameter equations the Freundlich equation (2) yielded a much closer fit to the sorption data than did the Langmuir equation (1).

Within the Langmuir family (1, 3, 4, 6) the proposed extended Langmuir equation (6) on an average yielded the closest fit followed by the four-parameter 'double' Langmuir (3), the three-parameter Gunary (4), and the two-parameter Langmuir (1) equations.

Within the Freundlich family (2, 5, 7) the proposed extended Freundlich equation (7) on average gave a slightly closer fit than the Fitter-Sutton equation (5). But on average both yielded a closer fit than any of the equations of the Langmuir family.

TABLE 2

The ability of seven mathematical equations to describe phosphate sorption data of eight soil samples; expressed as unexplained variation (residual sum of squares, RSS) relative to total variation (total sum of squares, TSS)

Sample No.	*Langmuir (1)*	*Double Langm. (3)*	*Gunary (4)*	*Propos. extend. Langm. (6)*	*Freundlich (2)*	*Fitter-Sutton (5)*	*Propos. extend. Freundl. (7)*
	RSS/TSS × 1000						
1a	22.2	0.26	0.37	0.13	1.21	0.11	0.10
1b	16.5	0.94	0.72	0.40	2.54	0.26	0.13
2a	23.2	0.65	0.78	0.37	0.52	0.28	0.32
2b	16.4	1.00	0.40	0.13	1.77	0.34	0.07
3a	13.9	0.32	0.15	0.16	1.04	0.22	0.18
3b	28.3	0.22	1.13	0.41	1.18	0.22	0.32
4a	23.9	0.66	0.81	0.30	0.30	0.29	0.30
4b	26.2	0.24	0.75	0.30	1.84	0.09	0.19
Mean of data set 1	21.3	0.54	0.64	0.28	1.30	0.23	0.20
Mean of data set 2	23.9	0.64	0.73	0.18	0.64	0.18	0.14

Data set 1: Total P sorbed = P sorbed experiment. + P extracted by resin.
Data set 2: Total P sorbed = P sorbed experiment. + 1.5 (P extr. by resin).

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Although not strictly appropriate for nonlinear models a partial F-test as used by Barrow (1978) may indicate whether an extension of an equation takes up a significant amount of variation over that removed by the original equation. Such tests showed that all three extended Langmuir equations (3, 4, 6) were significantly (99.9 per cent level) better than the simple Langmuir equation (1) on all soil samples. The four-parameter 'double' Langmuir equation (3) was significantly (95 per cent level) better than the proposed three-parameter Langmuir equation (6) only on soil sample 3b. The two extended Freundlich equations (5, 7) were significantly (99.9 per cent level) better than the simple Freundlich equation (2) on all samples except on the two FYM surface samples, 2a and 4a.

A detailed picture of the fit of the equations may be obtained from a plot of the residuals versus the concentration of P. A typical plot (Fig. 3) shows that the Freundlich equation (1) overestimated the P sorbed at low P concentration (negative residuals). It underestimated at medium P concentration and again overestimated at high concentration. The residuals of the Langmuir equation (2) showed the opposite pattern. The 'double' Langmuir equation (3) simply duplicated the residual pattern of the 'single' Langmuir equation (2). The residual patterns of equations 4, 5, 6 and 7 were similar to each other, but the Fitter-Sutton equation (5) gave the smallest amplitudes at low P concentration.

As stated above the P already present in the soil was estimated by anion exchange resin extraction. The rate of its desorption is different from the rate of the adsorption, and the amount of resin extractable P might be an underestimate of the labile pool. However, when the P in this pool in data

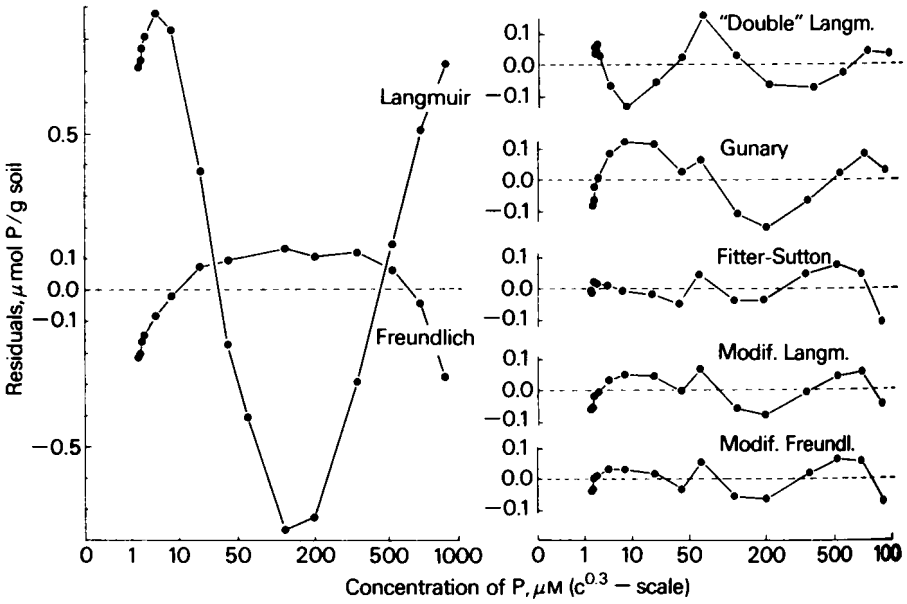

FIG. 3. Residuals obtained by fitting seven mathematical equations to phosphate sorption data of soil sample 1a.

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set 2 was increased to 1.5 times the resin extracted P, a poorer fit to the sorption data was obtained by the Langmuir (1), the 'double' Langmuir (3) and the Gunary (4) equations, but a closer fit was obtained by the Freundlich (2) and the two proposed three-parameter equations (6, 7). With the Fitter-Sutton equation (5) the residual sum of squares, RSS naturally did not change, as the constant D of this equation solely reflects the increase of the labile pool. The mean values of RSS relative to TSS of data set 2 (bottom of Table 2) show that the simple Freundlich equation (2) on average described the data just as well as the four-parameter 'double' Langmuir equation (3) and better than the Gunary equation (4). Furthermore, it may be seen that the proposed extended Freundlich equation (7) on average still yielded the closest fit to the sorption data.

In evaluating various sorption equations due consideration should also be given to the fact that the parameters of the equations as explained by Barrow (1978) may be more or less correlated. If two parameters of an equation are highly correlated, variation in one can be partly compensated by variation in the other, and the more care should be taken in interpreting the estimated parameters. It is another disadvantage of such equations that they often need relatively many iteration steps in the computing process of finding a minimum RSS.

Table 3 shows that the constants of the proposed extended Freundlich equation (7) were less correlated than the constants of the extended Langmuir equation (6) and especially less than the constants of Fitter-Sutton equation (5).

In general a mechanistic model for describing P sorption by soils is preferable to an empirical equation. A mechanistic model, however, has to be developed from an intimate physical and chemical knowledge of the behaviour of the phosphate ions in the solution-soil system. It is not sufficient to obtain a close fit of a postulated model with a series of observations (Bliss, 1970, Barrow, 1978.)

There is no unambiguous way to choose the 'best' equation as it depends on the purpose of the study. If the only aim is to classify soils with respect to their P sorption characteristics, both the Freundlich and the Langmuir

TABLE 3

Correlation between parameters within three sorption equations fitted to sorption data of eight soil samples

	Parameters paired		
	A – B	*A – D*	*B – D*
	*Asymptotic correlation coefficient**		
Fitter-Sutton (5)	−0.996	0.986	−0.972
Prop. extend. (6) Langm.	−0.958	−0.959	0.846
Prop. extend. (7) Freundl.	−0.928	−0.813	0.968

*Mean of coefficients obtained from eight soil samples.

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equations could be used, as the closeness of fit to the observations would then be of minor importance. But if the P sorption curve is to be used, for example, in a study of plant P uptake, stress would be put on the attainment of a close fit to the observations. In this respect, the proposed extended Freundlich equation would be the best as seen from the results presented here. Furthermore, the parameters of this equation were less correlated than the parameters of the other two well fitting equations (5, 6).

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